

Evaluation of Nonlinear Contrast Pulse Sequencing for Use in Super-Resolution Ultrasound Imaging

Katherine G. Brown¹, Student Member, IEEE, and Kenneth Hoyt², Senior Member, IEEE

Abstract—The use of super-resolution ultrasound (SR-US) imaging greatly improves visualization of microvascular structures, but clinical adoption is limited by long imaging times. This method depends on detecting and localizing isolated microbubbles (MBs), forcing the use of a dilute contrast agent concentration. Contrast-enhanced ultrasound (CEUS) image acquisition times as long as minutes arise as the localization of thousands of MBs are acquired to form a complete SR-US image. In this article, we explore the use of nonlinear CEUS strategies using nonlinear fundamental contrast pulse sequencing (CPS) to increase the contrast-to-tissue ratio (CTR) and compare MB detection effectiveness to linear B-mode CEUS imaging. The CPS compositions of amplitude modulation (AM), pulse inversion (PI), and a combination of the two (AMPI) were studied. A simulation study combined the Rayleigh–Plesset–Marmottant (RPM) model of MB characteristics and a nonlinear tissue model using the k-Wave toolbox for MATLAB (MathWorks Inc., Natick, MA, USA). Validation was conducted using an *in vitro* flow phantom and *in vivo* in the rat hind-limb. Imaging was performed with a programmable US scanner (Vantage 256, Verasonics Inc., Kirkland, WA, USA) and customized to transmit a set of basis US pulses from which both B-mode US (frame rate (FR) of 800 Hz) and multiple nonlinear CPS compositions (FR of 200 Hz) could be assessed from identical *in vitro* and *in vivo* datasets using a near simultaneous method. The simulations suggest that MB characteristics, such as diameter and motion, help to predict which US imaging strategy will enhance MB detection. The *in vitro* and *in vivo* US imaging studies revealed that different subpopulations of polydisperse MB contrast agents were detected by linear imaging and by each different nonlinear CPS composition. The most effective single imaging strategy at a 200-Hz FR was found to be B-mode US imaging. However, a combination of B-mode US imaging with a nonlinear CPS imaging strategy

was more effective in detecting MBs *in vivo* at all depths and was shown to shorten image acquisition time by an average of 33.3%–76.7% when combining one or more CPS sequences.

Index Terms—Contrast agents, contrast pulse sequencing (CPS), microbubbles (MBs), nonlinear ultrasound (US) imaging, super-resolution ultrasound (SR-US), US imaging.

I. INTRODUCTION

A VARIETY of pervasive and grave diseases, such as cancer, diabetes, and cardiovascular disease, induce changes in the microvasculature as they progress. Improved characterization of the microcirculation could better inform the diagnosis and treatment of these diseases [1], [2]. However, such a detailed characterization is limited with the use of current clinical imaging modalities, e.g., computed tomography (CT), magnetic resonance (MR), and ultrasound (US), due in part to spatial resolution limitations.

A technique known as super-resolution ultrasound (SR-US) has recently emerged for *in vivo* imaging of microvascular circulation as it offers a ten-fold improvement in spatial resolution over traditional US imaging methods [3]. Importantly, SR-US can achieve these spatial resolutions below the diffraction limit of US at clinically relevant depths [4]–[6]. This technique precisely localizes the centers of microbubble (MB) contrast agents from a time series of contrast-enhanced US (CEUS) images as the MBs flow throughout the microvasculature. Within only a few years, several groups have demonstrated that *in vivo* SR-US imaging is an effective new modality for investigating clinically relevant conditions like the properties of microvasculature structures [7]–[16] and cancer response to drug treatment [17], [18].

Clinical use of SR-US is presently limited by several factors that preclude its use as a real-time modality, including long image acquisition times and a heavy computational burden. These lengthy imaging times result in undesired motion in the image stream from respiration and other sources, thus creating an additional processing challenge. Image acquisition time is driven by the requirement to localize isolated MBs in a dilute solution. The dilute solution ensures that the signal from individual MBs do not overlap and prevent localization. Under these conditions, depicting the entire width of a vascular segment can require up to minutes to ensure a sufficient number of US image acquisitions and MB detection events.

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Katherine G. Brown is with the Department of Bioengineering, The University of Texas at Dallas, Richardson, TX 75080 USA (e-mail: katherine.brown@utdallas.edu).

Kenneth Hoyt is with the Department of Bioengineering, The University of Texas at Dallas, Richardson, TX 75080 USA, and also with the Department of Radiology, University of Texas Southwestern Medical Center, Dallas, TX 75235 USA (e-mail: kenneth.hoyt@utdallas.edu).

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Imaging for SR-US has been performed with CEUS using linear B-mode US [3], [12] or nonlinear US imaging using contrast pulse sequencing (CPS) compositions such as amplitude modulation (AM) [19], [20], pulse inversion (PI) [21], [22], or AMPI [17], [18], [23]. Imaging time is driven by both the signal-to-noise ratio (SNR) and contrast-to-tissue ratio (CTR) of the MB relative to its background, as these are the critical parameters for determining whether an MB will be detected in an US image frame. B-mode CEUS imaging has a high SNR but a limited CTR. Nonlinear CPS has been shown to take advantage of the nonlinear properties of the MB in an acoustic field to improve CTR relative to B-mode US imaging [24], [25].

The most commonly studied CPS compositions of PI, AM, and AMPI generally have found slight differences in CTR [26]–[28]. In PI, backscattered US signals from two pulses with 180° phase differences are combined to remove the linear portion of the signal associated with tissue. In AM, the amplitude of the US pulses transmitted is altered, and the backscattered US signals are subtracted to enhance the nonlinear content. For AMPI, both of these techniques are employed. However, nonlinear imaging requires additional pulses to be transmitted which lowers the maximum imaging frame rate (FR) achievable.

Reduction in the maximum FR with multipulse CPS imaging strategies leads to a tradeoff between FR and CTR. Additional pulse waveforms support CPS imaging strategies that raise CTR at the expense of FR. An additional factor is that FR is also impacted by the number of angles used in the compounding of plane wave images. More specifically, Couture *et al.* [25] showed that CTR increases linearly with the square root of the number of angles used in compounding. As shown by Christensen-Jeffries *et al.* [29], the product between the number of US pulses and compounded angles determines the FR.

In theoretical studies of the dynamic nonlinear behavior of an MB the popular Rayleigh–Plesset model has often been used. The Rayleigh–Plesset model of MB response in an US field was further extended by Marmottant to account for MB shell properties [30]. Predictions of MB response have been shown to match experimental observations by optical microscopy [31]–[33]. The Rayleigh–Plesset–Marmottant (RPM) model has also been used to understand the MB response to PI [22] and to AM and PI pulse compositions [19] used for SR-US imaging.

The motivation of the work presented herein was to increase the number of MBs detected and localized per imaging frame using linear and nonlinear US imaging strategies. The ultimate goal was to reduce the number of US frames required to visualize fully perfused microvascular structures as depicted in SR-US images. To that end, we evaluated the use of fundamental AM, PI, or AMPI nonlinear CPS compositions during CEUS imaging and detection of polydisperse MBs in circulation for SR-US image formation. In simulation, we employed an RPM model of an MB embedded in soft tissue. From these simulations, we predicted the CTR of an isolated MB. Next, we confirmed these simulation results with an *in vitro* flow phantom study. Finally, we evaluated the

proposed *in vivo* CEUS imaging methods in the hind limb of healthy rats. For the US imaging work, we used a near simultaneous method of forming the B-mode US and CPS images from a set of four basis pulses emitted in sequence that were linearly combined to form multipulse sequences. SR-US image formation during a subsequent processing step allowed a novel comparison method between linear and nonlinear CEUS imaging strategies based on an identical dataset for each. Overall, we show the combination of linear B-mode and nonlinear US imaging strategies can increase the number of MBs detected per frame and decrease the image acquisition time needed for SR-US image formation.

II. MATERIALS AND METHODS

A. Contrast Pulse Sequences

A study compared the nonlinear CPS compositions of AM, PI, and AMPI to the conventional linear pulsing strategy of B-mode US alone. All compositions were formed by combining one or more backscattered US signals from four pulses consisting of a full-amplitude pulse, an inverted full-amplitude pulse, and two half-amplitude pulses.

B-mode US images were formed from the positive full-amplitude basis pulse ($P_{1.0}$). Several alternative B-mode US compositions were formed from the inverted full-amplitude basis pulse ($P_{-1.0}$) and half-amplitude basis pulses ($P_{0.5a}$ and $P_{0.5b}$) and denoted as follows:

$$\text{B-mode} = P_{1.0} \quad (1)$$

$$\text{B-mode}_i = -P_{-1.0} \quad (2)$$

$$\text{B-mode}_{ii} = 2P_{0.5a} \quad (3)$$

$$\text{B-mode}_{iii} = 2P_{0.5b} \quad (4)$$

US signals were combined in the radio frequency (RF) domain from the basis pulses (prior to angular compounding) to form CEUS images according to the following equations:

$$\text{PI-2} = P_{1.0} + P_{-1.0} \quad (5)$$

$$\text{AM-2} = P_{1.0} - 2P_{0.5a} \quad (6)$$

$$\text{AM-3} = P_{1.0} - P_{0.5a} - P_{0.5b} \quad (7)$$

$$\text{AMPI-2} = P_{-1.0} + 2P_{0.5a} \quad (8)$$

$$\text{AMPI-3} = P_{-1.0} + P_{0.5a} + P_{0.5b} \quad (9)$$

Many of the above compositions have been studied and found to be effective in increasing CTR [28], [34]. Using a hydrophone and water tank setup, preliminary studies by our group investigated the use of different US pulse waveforms and additive compositions for the purpose of maximizing linear signal cancellation [35]. All demodulation was done at the fundamental frequency, including that for PI-2, which is more often demodulated at the second harmonic frequency. However, wideband US transducers that adequately span the center frequency and the first harmonic are required to effectively image with use of harmonic PI-2. PI-2 at the fundamental frequency was included for this reason and to better understand the individual contributions of both AM and PI to the performance of the commonly employed AMPI composition.

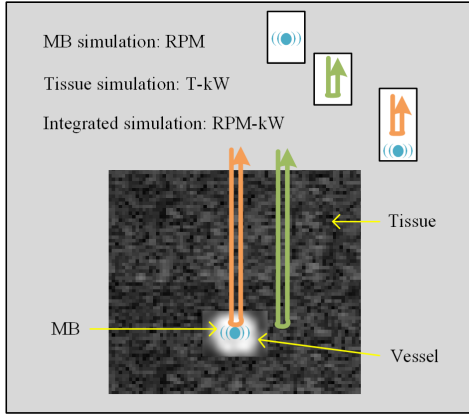


Fig. 1. Diagram of simulations based on the RPM model for the MB and k-Wave software for tissue. The round-trip tissue simulation is denoted T-kW, and the integrated MB and tissue simulations are RPM-kW.

B. Simulations

The overall flow of the integrated simulation environment for MBs and tissue is shown in Fig. 1, and is similar to that first described by Brown *et al.* [22]. Briefly, MB were studied using the RPM model and shell parameters consistent with the Definity contrast agent (Lantheus Medical Imaging, North Billerica, MA, USA) [36]. The simulation of the MB alone is referred to as the RPM simulation. The open-source k-Wave toolbox for MATLAB (MathWorks Inc., Natick, MA) was used to model the nonlinear propagation of US waves through tissue by solving coupled nonlinear differential equations, as detailed in [37]. The integrated simulations of the MB response with tissue use the propagation of an US pressure wave through a k-Wave tissue simulation to the depth of the MB where the waveform is captured. The captured waveform is used to stimulate the MB model, and to calculate the resulting US pressure wave. This pressure wave is then fed back into a second tissue simulation in kWave where the wave is propagated back through tissue to the transducer where the response is recorded. The integrated simulations for the MB and tissue are referred to as RPM-kW. Each simulation was repeated for the three basis pulses: full-amplitude pulse, inverted full-amplitude pulse, and single half-amplitude pulse. From the responses to the basis pulses, the nonlinear response to the two-pulse sequences of AM, PI, and AMPI were calculated. A round-trip tissue simulation in k-Wave, referred to as T-kW hereafter, used pressure waves with the same US driving pressures employed in the stimulus of the MB, propagated to the depth of the MB where they were backscattered from tissue and recorded by the transducer placed at the surface. The results of the T-kW simulation were used in calculation of a CTR as described in Section II-G.

1) **MB Simulations:** MBs were modeled by the RPM equations to account for the physical properties of the lipid MB shell [30]

$$\rho_l \left(R \ddot{R} + \frac{2}{3} \dot{R}^2 \right) = \left[P_0 + \frac{2\sigma_0(R_0)}{R_0} \right] \left(\frac{R}{R_0} \right)^{-3\kappa} \left(1 - \frac{3\kappa}{c} \dot{R} \right) - P_0 - \frac{2\sigma_0(R)}{R} - 4\mu_0 \frac{\dot{R}}{R} - 4\kappa_s \frac{\dot{R}}{R^2} - P_{ac}(t) \quad (10)$$

TABLE I
PARAMETERS USED FOR THE MB SIMULATION

Parameter	Value
Density (ρ_l)	1055 kg/m ³
Surface tension (σ_0)	0.07275 N/m
Dynamic viscosity (μ_0)	2×10^{-3} Ns/m ²
Speed of sound (c)	1540 m/s
Gas exponent (κ)	1.07
Shell viscosity (κ_s)	6×10^{-9} kg/s
Shell elasticity (χ)	0.81 N/m ²
Ambient pressure (P_0)	101325 kg/m ³

where ρ_l is density, σ_0 is surface tension, κ is the polytrophic exponent of the gas inside the MB, c is the speed of sound, μ_0 is dynamic viscosity, κ_s is the shell viscosity, P_0 is ambient pressure, P_{ac} is the acoustic pressure signal, R is the radius, and R_0 is the equilibrium MB radius. The effective surface tension term $\sigma_0(R)$ was modeled as a function of MB radius

$$\sigma_0(R) = \begin{cases} 0, & \text{if } R \leq R_{\text{buckling}} \\ \chi \left(\frac{R^2}{R_{\text{buckling}}^2} - 1 \right), & \text{if } R_{\text{buckling}} < R < R_{\text{ruptured}} \\ \sigma_{\text{water}}, & \text{if } R \geq R_{\text{ruptured}} \end{cases} \quad (11)$$

where χ is the MB shell elasticity. The MB buckling and rupture states were chosen as

$$R_{\text{buckling}} = \frac{R_0}{\left(1 + \frac{\sigma_0(R_0)}{\chi} \right)^{\frac{1}{2}}} \quad (12)$$

and

$$R_{\text{ruptured}} = R_{\text{buckling}} \left(1 + \frac{\sigma_{\text{water}}}{\chi} \right)^{\frac{1}{2}}. \quad (13)$$

MB shell parameters used in this study for Definity are derived from the work by Helfield and Goertz [36]. Other parameters used in the MB simulation are from Brown *et al.* [22]. All MB simulation parameters are summarized in Table I. Using (10), the time-dependent MB radius response was evaluated for six different diameters ranging from 1 to 6 μm .

The solution to (10) gives radial response (R), shell velocity ($\dot{R}$), and shell acceleration ($\ddot{R}$) values. From these, the US pressure P at a distance R_p from the MB can be determined [38]

$$P = \frac{\rho_l}{R_p} \left(R^2 \ddot{R} + 2R \dot{R}^2 \right) - \left(\frac{R}{R_p} \right)^4 \left(\frac{\rho_l \dot{R}^2}{2} \right). \quad (14)$$

The pressure response was determined at a distance of $R_p = 50 \mu\text{m}$ from the MB and then rescaled to produce a matching backscattered US signal that propagates through tissue using a k-Wave simulation.

2) **Tissue and Transducer Simulations:** Heterogeneous tissue was modeled as a 3-D block that was $20 \times 10 \times 4.5 \text{ mm}^3$ in size (depth $\times$ width $\times$ elevation) on a grid of points that were uniformly spaced by 0.04 mm, with a mean scatterer density of 0.3% and standard deviation of 0.8% of the bulk

TABLE II
PARAMETERS USED FOR THE k-WAVE SIMULATION

Parameter	Value
Tissue density	1050 kg/m ³
Tissue speed of sound	1570 m/s
Blood density (ρ)	1060 kg/m ³
Blood speed of sound (c)	1584 m/s
Power law attenuation coeff. (α)	1 dB/MHz/cm
Power law attenuation exponent (γ)	1.05
B/A nonlinear exponent	7.4

properties. In this tissue block, an area of inhomogeneity was embedded into the otherwise homogeneous block to model a blood vessel of radius 1 mm at a depth of 6 mm. The tissue properties used in the k-Wave simulation are summarized in Table II [22], [39]. For simulation purposes, the MB location was centered in the blood vessel.

The US transducer model consisted of 32 elements transmitting a plane wave at a single angle and having Hamming window apodization on both transmit and receive. The input pressure waveform was modeled as a Gaussian US pulse of two cycles in length with a center frequency of 6.25 MHz and a mechanical index (MI) of 0.1. A filter was constructed to closely match the sensitivity of the transducer used in the experimental study and was applied to all waveforms received by the transducer. MB motion was modeled using the k-Wave toolbox by fixing the MB but imposing tissue motion in the lateral and axial directions [22], each by a one grid element (0.04 mm) to simulate a speed of approximately 11 mm/s given an effective FR of 200 Hz.

C. US Imaging

Imaging was performed using an ultrafast plane wave technique with a programmable US scanner (Vantage 256, Verasonics Inc., Kirkland, WA, USA) equipped with an L11-4v linear array transducer. As in simulation, transmit pulses had a length of two cycles with a center frequency of 6.25 MHz, and an MI for a full-amplitude pulse of 0.1. A half amplitude pulse was formed by adjusting the amplitude of the pulse waveform transmitted by the hardware and was confirmed by a hydrophone test (AIMS III, Onda Corporation, Sunnyvale, CA, USA) with the transducer in water tank setup [35]. Measured waveforms and signal cancellation levels when combined as AM, PI, and AMPI compositions are shown in Fig. 2. The order of US pulses for each of the five transmit angles (-4° , 2° , 0° , 2° , 4°) was a half-amplitude pulse, a full-amplitude pulse, an inverted full-amplitude pulse, and a second half-amplitude pulse using a pulse repetition frequency of 7.1 kHz. The FR for the individual US imaging strategies studied was set at 200 Hz. Time gain compensation (TGC) settings were fixed and the same for all US imaging sessions. The RF signals were stored for offline analysis using custom MATLAB software (MathWorks, Inc.).

D. In Vitro Experiments

A flow phantom was constructed from a 10% gelatin and 1% scatterer mixture (by weight/volume) in water solution [40].

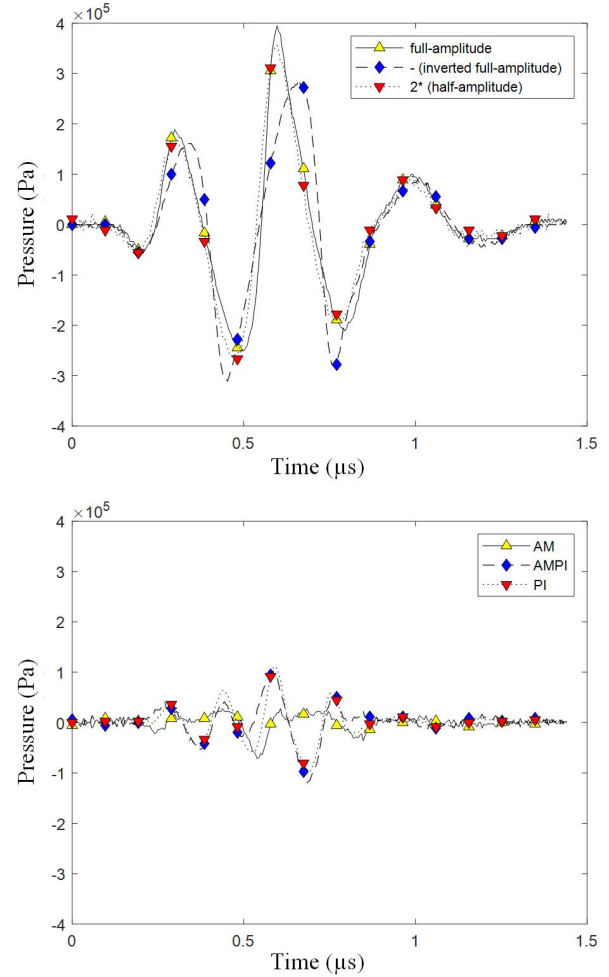


Fig. 2. Measured US pulse waveforms measured for full-amplitude, inverted full-amplitude, and half-amplitude (left) pulses, showing slight differences in waveform rise and fall times of the scaled amplitudes. CPS compositions (right) of AM, PI, and AMPI from the measured pulse waveforms depict the level of cancellation in a water medium.

This mixture was heated to 50 °C and then poured into a rigid mold threaded with a Teflon wire. After 8 h of refrigeration and phantom solidification, the wire was removed leaving a hollow void representative of a small vessel. The channel had a diameter of 2.25 mm at a depth of 17 mm. A commercial contrast agent (Definity, Lantheus Medical Imaging), was diluted to 4×10^6 times the original concentration, derived empirically, to improve the isolation of MB in solution. This MB solution was slowly pumped through the phantom at pulsatile flow rates of 8–10 mL/min (Model 77200-60, Cole-Palmer, Vernon Hills, IL, USA). A total of 7200 frames were captured in 48 blocks of 150 frames over 3.5 min. A total of five independent datasets were acquired at the different flow rates.

E. In Vivo Experiments

These studies were performed on the hindlimb of healthy male Sprague Dawley rats ($N = 5$; Charles River Laboratories, Wilmington, MA). A 12.5- μ L bolus of contrast agent (Definity, Lantheus Medical Imaging) diluted in 12.5- μ L saline was

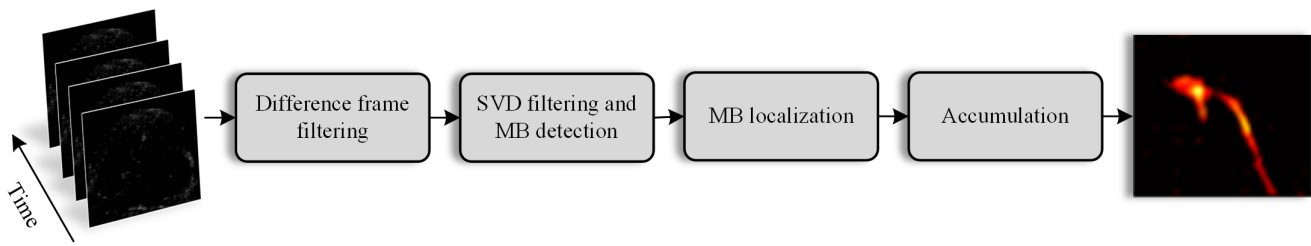


Fig. 3. Image processing steps used to produce an SR-US image. MB = microbubble and SVD = singular value decomposition.

slowly injected in the tail vein of anesthetized animals. With the transducer fixed over the tissue area of interest, US imaging began as the MB bolus was administered and continuously for 3.5 min thereafter while capturing 7200 frames. All animal experiments were approved by the Institutional Animal Care and Use Committee (IACUC) at The University of Texas at Dallas, Richardson, TX, USA.

F. SR-US Image Processing

The SR-US image processing steps are illustrated in Fig. 3. Image formation begins with tissue suppression, followed by MB centroid localization and enumeration [7]. Datasets in this study displayed minimal motion (less than 50 μm) as imaging was performed in the secured hindlimb. Thus, no motion correction techniques were used. Tissue suppression involved use of a finite impulse response (FIR) difference filter $[-1 \ 1]$ that subtracted successive frames of the US image stack. Additional tissue suppression was implemented with singular value decomposition (SVD) filtering [41]. Briefly, SVD filtering was accomplished by removing the lowest order (highest magnitude) singular values (SVs) representing tissue from a sequence of US images. The SV cutoff varied for the linear and nonlinear US pulsing strategies, with a greater tissue signal in the linear images requiring a higher SV cutoff [42]. The SV threshold for removal was determined experimentally by analyzing the percent saturation of MBs filling the diameter of selected vessels in the SR-US images and by qualitative analysis. Filtering was computed independently for the three 2400-frame image substacks in the 7200-frame sequence for each of the linear and nonlinear US imaging strategies. Note that as statistics of the US images were different for each imaging strategy as well as for the first, second, and third substacks for the same strategy, optimal SV cutoff values were independently determined for each.

The SVD filtered images were spatially upsampled to their final resolution level of $\lambda/8$. Precise MB centroids in the image stacks were obtained by convolution with a 2-D Gaussian kernel fit to the measured point spread function (PSF) of the US system [23]. Gaussian kernels fitted to the PSFs for each of the imaging strategies were found to have very slight differences in axial and lateral dimensions so an average value was used. MBs detected with overlapping PSFs were rejected to avoid potential bias. Localizations were enumerated across all frames to form the final SR-US images.

Several combined linear and nonlinear US imaging strategies were studied in order to maximize the number of MBs

detected per frame by taking advantage of their unique potential to detect different MB subpopulations. For each strategy added in forming a combined strategy, there is a proportional increase in processing time when using conventional methods. Due to the increase in processing rate burden, two simple “Top-2” strategies were also considered; one approach combined two of the most advantageous linear strategies and the other combined B-mode US with the most advantageous nonlinear strategy. To combine nonlinear strategies within a 2400 frame subset, or the full 7200 frames, the logical union of MB localizations was performed. This had the effect of minimizing any repeat counting of the same MB that might arise in multiple strategies. Combined linear strategies were formed from combining US images based on raw pulses, scaled as in (1)–(4). Five combined strategies were evaluated

Top-2BC: B-mode + AMPI-3

Top-B400: B-mode + B-mode_i

C: AMPI-3 + AM-2 + AM-3 + PI-2

B800: Bmode + B-mode_i + B-mode_{ii} + B-mode_{iii}

B800&C: B800 + AMPI-3 + AM-2 + AM-3 + PI-2

where Top-2BC refers to a combination of the individual method with the highest measured saturation (B-mode US) and the nonlinear method with the highest measured saturation (AMPI-3). Based on *in vitro* results, Top-B400 refers to a pseudo-B-mode US 400-Hz FR combined image formed with the two linear strategies found to detect the most MBs (from the +1.0 and −1.0 pulses). Similarly, B800 refers to a linear pseudo-B-mode US 800-Hz FR combined image formed with all four pulses, where C (denoted “C” for CPS) refers to a combination of all nonlinear strategies, and B800&C refers to a combination of B800 (all linear imaging strategies) with all nonlinear US imaging strategies.

Defined as the number of US image frames to be processed per sec, the processing rate burden for each US imaging strategy examined in this study is summarized in Table III.

G. Quantification

A relative CTR (CTR_R) was used to compare results of linear and nonlinear US imaging strategies in simulation [43]. Using (5)–(8), the combined acoustic pressure responses of the MB in the simulation to the applied basis pulse waveforms as input were formed for each CPS composition. These were compared to that of the MB pressure response to the

TABLE III
IMAGE PROCESSING RATE BURDEN

Imaging strategy	Rate (Hz)
Individual strategies (e.g. B-mode US, AMPI-3, etc.)	200
Top-2BC	400
Top-B400	400
C	1,000
B800	800
B800&C	1,800

basis pulse representing a B-mode US stimulus as a relative metric, CTR_R . Note that a value of 0 dB would indicate equivalence and a negative value would indicate a lower CTR than B-mode US pulsing. General equations for these terms are

$$CTR = \frac{\mu_{\text{bubble}}}{\mu_{\text{tissue}}} \quad (15)$$

$$CTR_R = 20 \log \left(\frac{CTR_{\text{CPS}}}{CTR_{\text{B-mode}}} \right) \quad (16)$$

where μ is the root mean squared (rms) of the acoustic pressure response measured prior to tissue suppression.

The equation for CTR_R was rearranged as follows for ease of computation:

$$CTR_R = 20 \log \left[\frac{\left(\frac{\mu_{\text{bubble}}}{\mu_{\text{tissue}}} \right)_{\text{CPS}}}{\left(\frac{\mu_{\text{bubble}}}{\mu_{\text{tissue}}} \right)_{\text{Bmode}}} \right] \quad (17)$$

$$= 20 \log \left[\frac{\left(\frac{\mu_{\text{bubble}}}{\mu_{\text{tissue}}} \right)_{\text{CPS}}}{\left(\frac{\mu_{\text{bubble}}}{\mu_{\text{tissue}}} \right)_{\text{Bmode}}} \right] \quad (18)$$

where the numerator is calculated with the MB model, and the denominator comes from a k-Wave simulation of a round trip backscattered US signal in tissue (T-kW). The denominator represents a tissue suppression term, as it is a measure of tissue signal cancellation for the AM, PI, and AMPI-based US imaging relative to B-mode US imaging.

A vessel saturation measure, expressed as a percentage, was defined within a region-of-interest (ROI) as follows:

$$\text{Saturation} = \frac{\sum_{\text{ROI}} (\text{pixel intensity} > 0)}{\sum_{\text{ROI}} (\text{total pixels})} \times 100 \quad (19)$$

where the pixel intensity is the sum of the MB localizations at that pixel.

A measure of the ratio of distinct MB detections with imaging strategy “A” with respect to strategy “B” was computed to reveal how different US imaging strategies might be combined to increase the total number of MBs detected per image frame. MBs from two strategies were paired with the Hungarian algorithm [44] on a frame-by-frame basis, using a maximum pairing distance based on the maximal lateral and axial extent of the PSF and taking into account motion. MBs that were paired were assumed to be the same MB with

potentially slightly different localizations due to the acquisition time difference of multiple pulses

$$\text{Distinct MB}_{AB} = \frac{MB_A - \text{Paired MB}_{AB}}{MB_B} \times 100 \quad (20)$$

where MB_A is the total MBs detected in a vessel ROI with imaging strategy “A,” and Paired MB_{AB} is the number of MBs that are paired in identical frame numbers for both US imaging strategies “A” and “B.”

III. RESULTS

A. Simulations

A lipid-shelled MB response to a sequence of US basis pulses was simulated using the RPM model with idealized waveforms and a range of steady-state MB diameters. The US pressure of the MB response to the basis pulses in Fig. 4 are depicted in such a way to highlight their asymmetries. Plots show the full-amplitude, negative of the inverted full-amplitude, and twice the half-amplitude pulses. These plots help highlight differences in the phase of each response for an MB with a diameter of 2 μm , which is a representative of the size of Definity (mean diameter range 1.1–3.3 μm). The nonlinear MB pressure response calculated by (14) in the RPM simulation is apparent from the waveform distortions [Fig. 4(a)]. Also, the half-amplitude response leads the full-amplitude response. The inverted full-amplitude response leads the full-amplitude signal in the negative half of the waveform but lags in the positive half of the waveform [see inset in Fig. 4(a)]. However, in the RPM-kW simulation results without motion [Fig. 4(b)] and with motion [Fig. 4(c)], both the inverted full-amplitude response and the half-amplitude response lead the full-amplitude response. The amplitude of an acoustic pressure response to CPS pulsing is increased when combining basis pulse responses with phasing differences. Thus, the effect of motion is expected to enhance the CTR for nonlinear US strategies.

The acoustic pressure responses to the three pulses (in Fig. 4) of an MB with an initial diameter of 2 μm were combined to form three CPS compositions as depicted in Fig. 5(a). The waveform exhibits frequencies other than the fundamental frequency which is a characteristic of MB resonance behavior.

The responses for RPM-kW in Fig. 5(b), (without motion), and Fig. 5(c) (with motion), are more attenuated and smoother as a consequence of the propagation of the US signals through a nonlinear tissue simulation and the application of a filter matching the sensitivity of the transducer.

At an US imaging frequency of 6.25 MHz, the RPM simulation suggests an advantage of nonlinear CPS compositions for diameters less than 3 μm (CTR_R values greater than 0 dB), and an advantage for the use of linear B-mode US imaging for MBs with diameters larger than 4 μm [Fig. 6(a)]. Over the simulated range of diameters, the CTR_R of the response to the PI sequence is higher than that for AM and AMPI, albeit, the differences are small. The CTR_R results for the integrated RPM-kW simulations without motion [Fig. 6(b)] and with motion [Fig. 6(c)] preserve the overall trends of the results of the stand-alone RPM simulation [Fig. 6(a)] as the

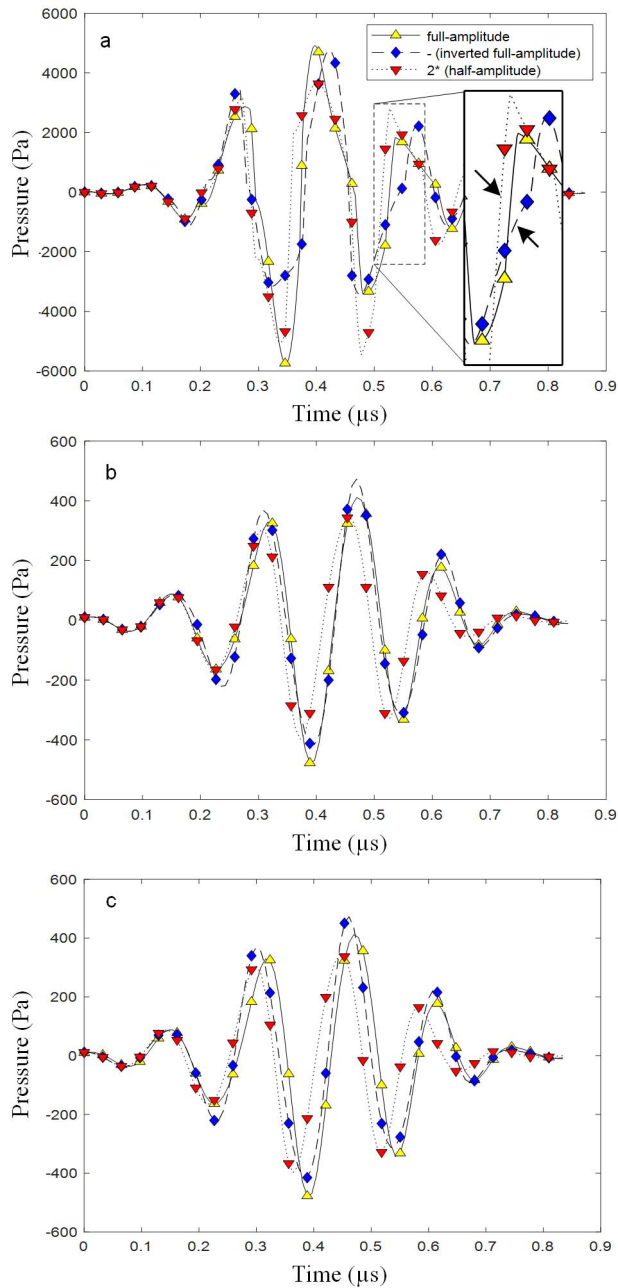


Fig. 4. Acoustic pressure responses of full-amplitude, inverted full-amplitude, and half-amplitude pulses from MB simulations with an initial diameter of $2 \mu\text{m}$ for the (a) RPM, (b) RPM-kW without motion, and (c) RPM-kW with small axial and lateral motion. Arrows in (a) point to the differences in phase of the inverted full-amplitude and half-amplitude waveforms with respect to the noninverted one. Note that the vertical scale is different in (a) than in (b) and (c).

tissue simulations are incorporated. While the results for the AM, AMPI, and PI sequences are tightly grouped, the AM sequence was a slightly higher CTR_R . In addition, the motion of the MB in Fig. 6(c) has a noticeable beneficial effect to the CTR_R for the responses to AM and PI sequences.

Depicted in Fig. 7(a), the full spectral response from the RPM simulation reveals harmonics generated at twice the fundamental frequency (i.e., 12.5 MHz), which are particularly

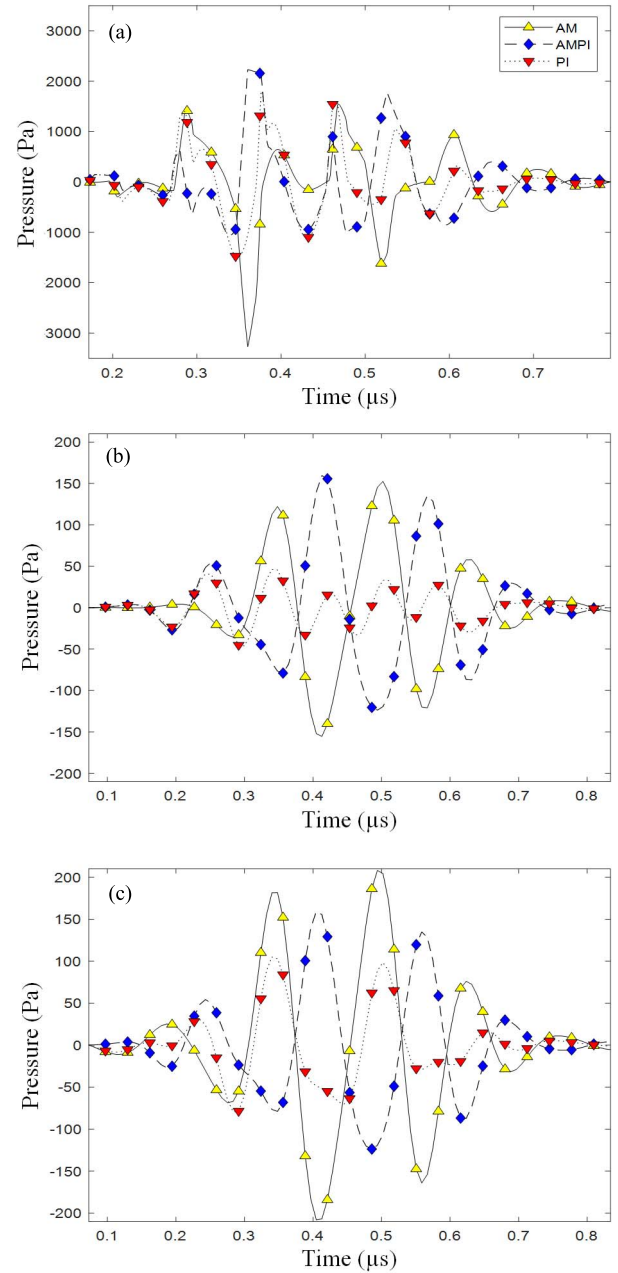


Fig. 5. Acoustic pressure responses of AM, PI, and AMPI pulse sequences from simulations at an initial diameter of $2 \mu\text{m}$ for the (a) RPM, (b) RPM-kW without motion, and (c) RPM-kW with small axial and lateral motion. Note that the vertical scale is different in (a) than in (b) and (c).

strong for the smaller diameter MBs and nonlinear US imaging strategies. This aspect of the nonlinear MB response is visible in waveforms with abrupt changes in slope indicative of the higher frequency content seen in the RPM simulation [Fig. 5(a)]. The spectral response for RPM-kW simulations [Fig. 7(b) and (c)] shows similar harmonic content but at a reduced level. This reduction is understood to be from US propagation through tissue as well as filtering by the sensitivity profile of the transducer. Both of these tend to attenuate higher frequencies. However, with motion added [Fig. 7(c)], there

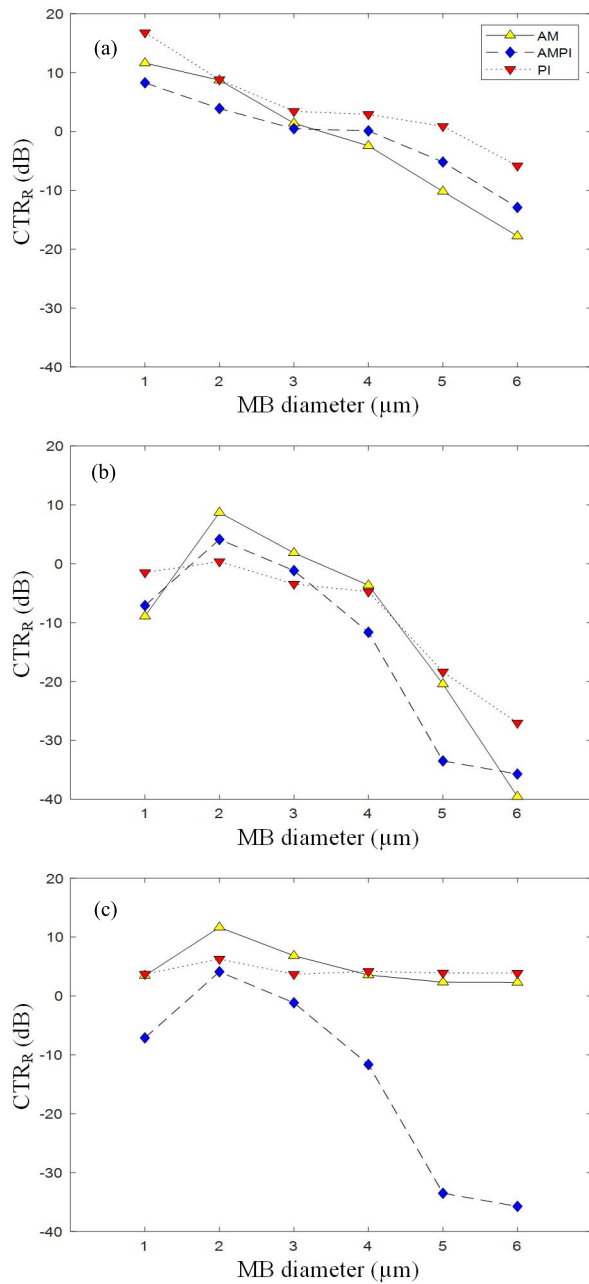


Fig. 6. CTR_R results from simulations of AM, PI, and AMPI pulse sequences at six diameters for the (a) RPM, (b) RPM-kW without motion, and (c) RPM-kW with small axial and lateral motion.

is an increase in the energy at the fundamental frequency (6.25 MHz) for both AM and PI pulsing, in concurrence with an increased CTR_R as seen in Fig 6(c).

B. In Vitro Results

In vitro experiments with a polydisperse MB contrast agent (Definity) in a flow phantom were carried out to explore whether the differences in CTR for linear and nonlinear US imaging strategies seen in simulation could be confirmed. In a detailed examination of SR-US localizations and comparison to intermediate processing results, it was determined that some MBs that were clearly visible and localized based on B-mode

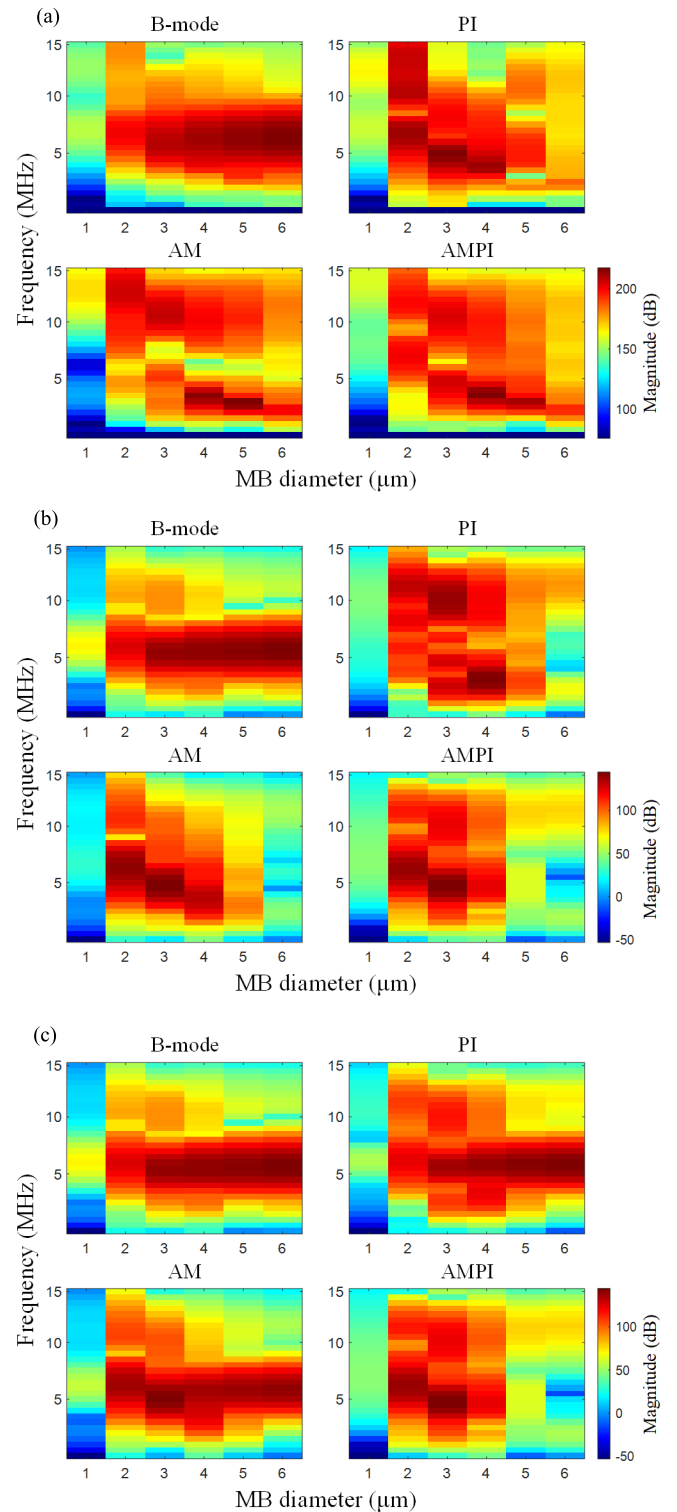


Fig. 7. Spectral results from simulations using the (a) RPM, (b) RPM-kW without motion, and (c) RPM-kW with small axial and lateral motion models for MBs with a range of diameters.

US imaging were not visible, nor were they localized, in one or more nonlinear US imaging strategies, and vice versa. Fig. 8 shows a comparison of the intermediate frame result (after the difference filter step in the SR-US processing flow) for four different US pulsing strategies. Each image depicts

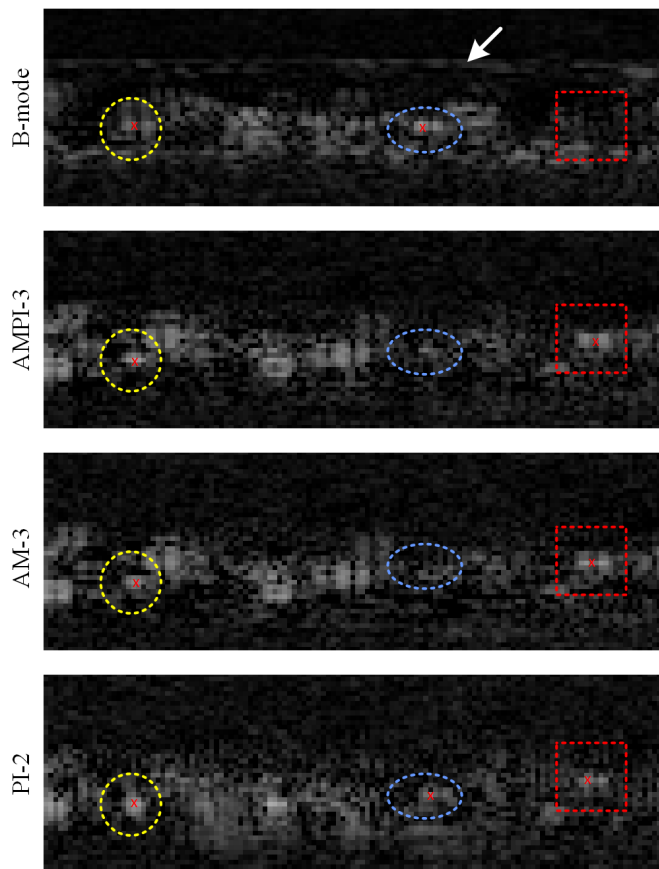


Fig. 8. Images from an *in vitro* US experiment produced with a near simultaneous method showing MB flowing in a channel with linear B-mode US, and nonlinear AMPI-3, AM-3, and PI-2 pulsing strategies. Qualitatively, the MB in the blue oval is more clearly depicted with B-mode US imaging, while the MB in the yellow circle with use of PI-2 pulsing and that in the red square using AMPI-3 pulsing. MB localizations are shown with a red “x,” and are missing in some images where the MBs were not detected.

a cropped portion of the flow channel for the same frame and highlights regions where MBs appear with varying CTR depending on the US imaging technique. Qualitatively, MBs in one region appear to have the highest CTR using B-mode US imaging. However, MBs in different locations have the highest CTR using PI-2 pulsing or are better visualized with AMPI-3 pulsing. In short, this demonstrates that a MB that is selectively detectable with one particular US imaging strategy may not be detected by another. Further, these differences in CTR led to differences where MBs could be detected in each frame of the SR-US stack using the different strategies. Overall, these *in vitro* experiments revealed that the use of a linear US imaging strategy detected more MBs within the vessel than any nonlinear imaging approach, but it failed to detect certain MBs that were detected by nonlinear US imaging strategies.

The *in vitro* study showed a qualitatively greater number of false positive MB detections with B-mode US compared to AMPI-3 imaging (see Fig. 9). Further, *in vitro* experiments revealed a qualitatively greater number of false positive MB detections in brighter tissue when using B-mode US imaging for SR-US compared to any nonlinear US imaging strategy.

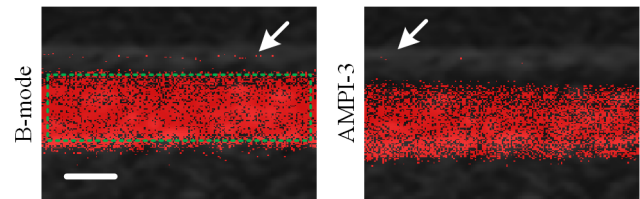


Fig. 9. SR-US images (MB detections are red dots) overlaid on B-mode US images (background) from an *in vitro* phantom experiment using B-mode US or AMPI-3 imaging for SR-US. Arrows point to false detections of MB within the upper wall of the flow channel with SR-US results, more prevalent for the B-mode US imaging strategy. The vessel ROI is outlined in green. Scale bar is 1 mm.

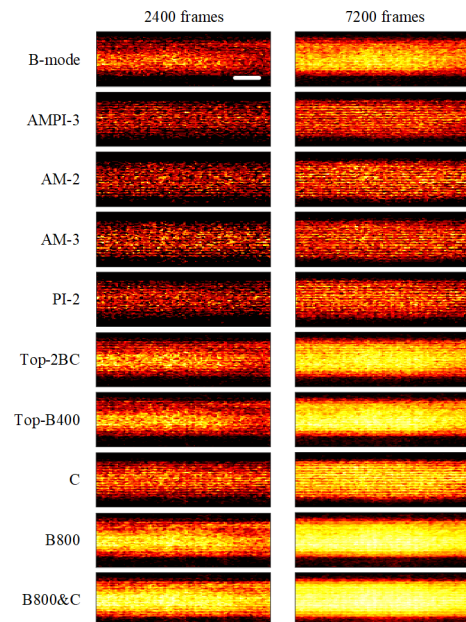


Fig. 10. SR-US images from the *in vitro* phantom experiment from 2400 (left) and 7200 (right) frames for linear, nonlinear, and combined US imaging strategies. Scale bar is 2 mm.

These experiments showed that a combined strategy increases the total brightness of the image and reflects a higher number of detected MBs as compared to individual linear or nonlinear US imaging strategies (see Fig. 10). Notably, the combined strategies making use of just 2400 frames (e.g., B800, B800&C) achieved a similar brightness and vessel fill-in as the individual US strategies using the full 7200 frames available (e.g., B-mode US, AMPI-3) while taking one-third of the acquisition time.

Table IV details the percentage of distinct MB_{AB} found by each individual US pulsing method when compared to each of the others in the *in vitro* study. From this measure it can be estimated how many additional MBs would be detected if method “A” were combined with method “B” (as a percentage). For instance, the top row indicates how many more MBs would be detected by additional methods combined with B-mode US. For example, the addition of the AMPI-3 imaging strategy to a B-mode US imaging strategy would on average detect 25.1% more MBs. This is better than other CPS strategies and is the rationale for use of the Top-2BC combined

TABLE IV
DISTINCT MB_{AB} RATIO (%)

A \ B	B-mode	AMPI-3	AM-2	AM-3	PI-2
B-mode	--	<u>25.1 ± 15.6</u>	20.4 ± 14.9	20.9 ± 14.0	24.1 ± 20.3
AMPI-3	41.8 ± 13.2	--	22.0 ± 7.0	22.5 ± 5.7	28.6 ± 1.02
AM-2	39.8 ± 12.2	26.4 ± 4.1	--	20.0 ± 9.3	50.5 ± 6.9
AM-3	40.7 ± 11.9	26.6 ± 3.6	18.8 ± 6.8	--	47.7 ± 7.6
PI-2	39.8 ± 13.2	30.7 ± 3.9	45.3 ± 4.5	44.1 ± 2.6	--

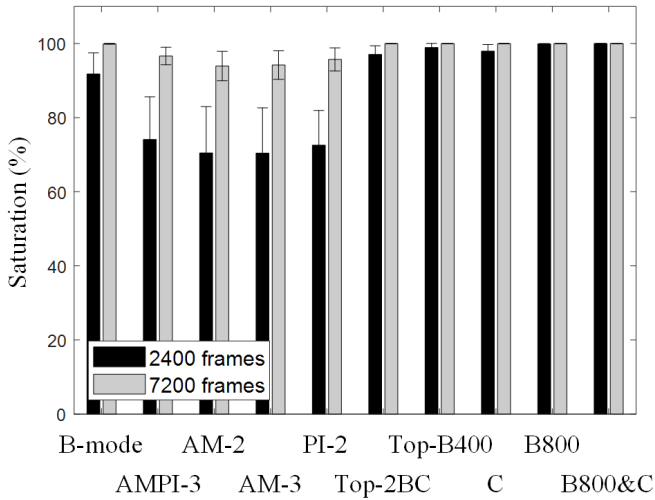


Fig. 11. Saturation of an ROI within the fluid channel of an *in vitro* phantom for SR-US with linear, nonlinear, and combined US imaging strategies using 2400 and 7200 frames. Top-2BC = B-mode US and AMPI-3, Top-B400 = B-mode and Bmode_i, C = all individual CPS compositions, B800 = pseudo-B-mode US at 800Hz, B800&C = pseudo-B-mode US at 800 Hz, and all individual CPS compositions.

US imaging strategy. For AMPI-3, the best advantage appears to be with combination with a B-mode US imaging strategy, while for AM-2 and AM-3 imaging, the best choice appears to be combining with PI-2 imaging, and vice versa. Separately, we found that the in pairing of MBs from US imaging with two identical half-amplitude pulses 20% of MBs were distinct. Additional MB detections may arise as a result of the time diversity of US pulsing methods composed of different basis pulses transmitted 140 μ s apart, or potentially from differences in CTR of the subpopulations of MBs under different US imaging strategies.

In vitro saturation of a vessel ROI with the use of linear, nonlinear, and combined imaging US strategies are compared in Fig. 11. The y-axis depicts the measured saturation of an ROI within the fluid channel. The bars indicate the relative saturation when using 2400 and 7200 frames. Of the noncombined strategies, B-mode US imaging resulted in the highest saturation levels. AMPI-3 imaging produced the second highest levels, which was only slightly higher than the levels reached by the other nonlinear US imaging strategies. All combined strategies exceeded the individual strategies, with B800, and B800&C filling 100% of the vessel after use

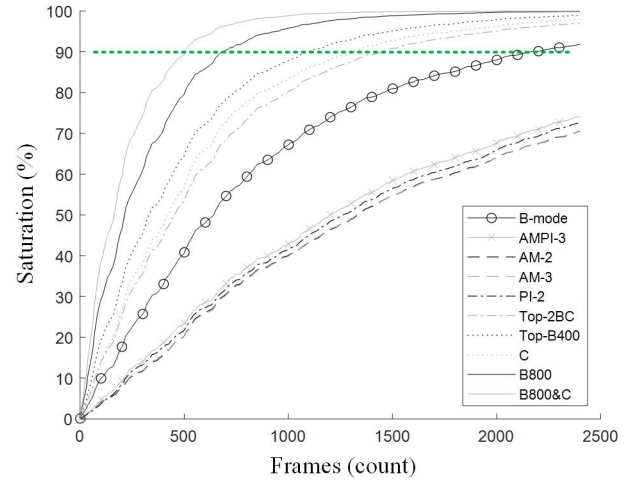


Fig. 12. Saturation curves from the *in vitro* phantom experiments for SR-US linear, nonlinear, and combined imaging strategies. Note that the curves for AM-2 and AM-3 overlay each other. The 90% saturation level is marked with a green dashed line for comparison of imaging strategies.

of 2400 frames. The simplest combined strategies, Top-2BC and Tob-B400, nearly achieved complete vessel fill-in.

Average saturation curves reveal that of the individual strategies, B-mode US imaging reaches a level of 90% saturation at 2166 frames, well before any single nonlinear US imaging strategy (see Fig. 12). The combined strategies reduce the number of US image frames to reach 90% saturation, with Top-B400 (at 1093 frames) reducing the time by an average of 49.5% and Top-2BC (at 1445 frames) by an average of 33.3%. The B800 and the C combined strategies reached image saturation at 692 and 1314 frames, reducing acquisition time by an average of 68.1% and 39.3%, respectively. The combination of all US imaging strategies, B800&C, reached 90% the earliest (at 504 frames), which would represent a reduced image acquisition time of 76.7%, but with considerably more processing required.

C. In Vivo Results

In vivo studies were performed in the healthy rat hindlimb to confirm the CTR results for linear, nonlinear, and combined imaging strategies from the *in vitro* experiments and in simulations. The *in vivo* results based on B-mode US and AMPI-3 imaging using 2400 and 7200 frames reveal slight

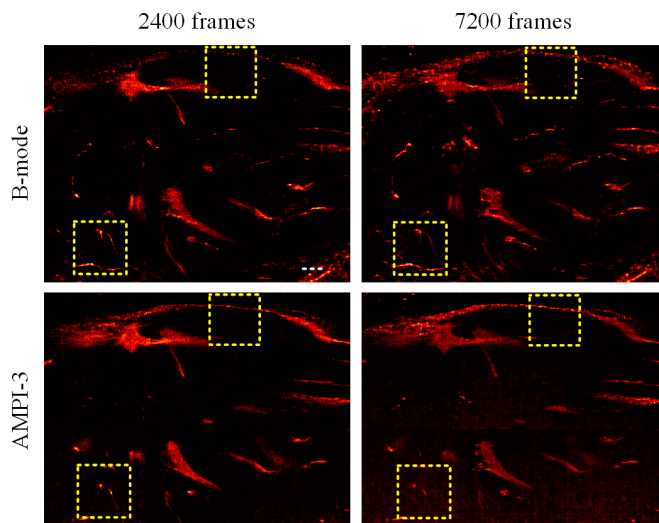


Fig. 13. SR-US images of the *in vivo* rat hind-limb using B-mode US and AMPI-3 imaging. Shallow vessels and deeper vessels (in boxed areas) are depicted with qualitatively different levels of detail. Scale bar is 2 mm. All figures are at the same scale.

image differences in the detail of the microvasculature. There were commensurately more MBs detected in 7200 frames than in 2400 frames for each strategy. There were qualitatively more MBs detected by the AMPI-3 strategy at shallow depths and more vessels delineated deeper by B-mode US imaging alone (see Fig. 13). However, there is an increased amount of noise with depth evident in the image from AMPI-3 using 7200 frames. While present, this effect is less apparent with B-mode US imaging using 7200 frames and is less apparent with AMPI-3 in the first 2400 frames as the MBs wash-in and the total signal MB strength is peaking.

SR-US images employing combined linear and nonlinear strategies in the *in vivo* study using just 2400 image frames are compared in Fig. 14. The comparison suggests that a combined linear and nonlinear US imaging strategy with modest additional processing, Top-2BC, fills both shallow and deep vessels adequately for visualizing form and measurement. The Top-B400 strategy shows slight differences in the depiction of shallow vessels compared to Top-2BC. The other three combined strategies also perform well using 2400 frames, though these require more processing resources. The B800 strategy reveals a slight weakness of B-mode US imaging for visualizing shallower structures while it is effective at tissue depth.

IV. DISCUSSION

This study evaluated the use of linear and nonlinear US imaging strategies to improve the detection of MBs during SR-US image formation. We first explored the effect of pulse sequences in simulation and then sought to confirm those results using a series of *in vitro* and *in vivo* studies. Couture *et al.* [6] recently summarized the linear and nonlinear pulse sequences for use during SR-US imaging and highlighted the potential of dedicated sequences to improve MB detection and localization. We found that a combination of linear and nonlinear US imaging strategies is most effective in detecting all subpopulations of MBs that differ in diameter

and velocity. The improvements in MB detection were shown to decrease the US imaging time by 33.3%–76.7% when producing an image with an intensity saturation level of 90% in a vessel of interest.

Simulation results revealed that the CTR has a dependence on MB diameter with the use of nonlinear US pulsing strategies. Of note, smaller diameter MBs were associated with increased CTR_R for nonlinear US pulsing strategies (Fig. 6). A similar result was noted by Newsome *et al.* [45] who studied the harmonic content of size-isolated MBs and found that smaller diameter MBs had higher nonlinear US signal content. The motion of the MBs also increased the CTR for nonlinear US pulsing strategies. These findings are consistent with the studies of Tremblay-Darveau *et al.* [19] who used the phase lag from AM pulsing to segment the MB from tissue signals. Their findings for both AM and PI are consistent with ours in that shell-softening of the MB induces a phase difference in the backscattered US signals of half-amplitude and inverted full-amplitude pulses (Fig. 4). This phase difference was predictive of the highly nonlinear response of smaller diameter MBs when imaged using US pulses with wavelength near the MB resonance frequency. Our results are also consistent with the RPM and k-Wave simulation of Brown *et al.* [22] on PI, SVD, and differential imaging (DI). While this group only looked at one nonlinear US pulsing strategy, they found that PI with harmonic imaging had a high CTR that was relatively independent of MB motion direction and speed and was thus, appropriate for slow flow and tortuous vascular network conditions.

It was also noted in simulation that smaller diameter MBs have a lower amplitude pressure response as compared to larger MBs, indicating a higher SNR for the latter. Of note, the rate of change of acoustic pressure response with diameter was greater under B-mode US imaging than under nonlinear imaging strategies. This is understood to come from the greater volumetric changes that larger MBs undergo, which is consistent with previous findings [22]. Thus, while the CTR of CPS images relative to B-mode US images of smaller MBs may be higher, the SNR is lower. It is understood that there is an interplay between SNR and CTR in the effective detection of MBs, and this was further examined in the *in vitro* and *in vivo* experiments.

The MB simulations used the shell properties of Definity [36]. The shell elasticity and viscosity parameters were chosen as an average over MB diameters. However, that study showed that there is a spread of MB values with diameter. A range of MB shell parameters were used within the experimental results from [36] and were found to have a small influence on results, and did not affect the general trends uncovered. The work by Faez *et al.* [46] examined fresh and expired Definity MBs and found that shell viscosity increased with aging, and the size distribution changed from a bimodal distribution with peaks at 1.5 and 4 μm to a single mode distribution peaking at a diameter of 2 μm . The dissolution of larger MBs and prevalence of smaller diameter MBs in older Definity contrast agent samples may suggest an advantage to using nonlinear US pulsing strategies, thereby increasing the CTR of the MBs in that case. While

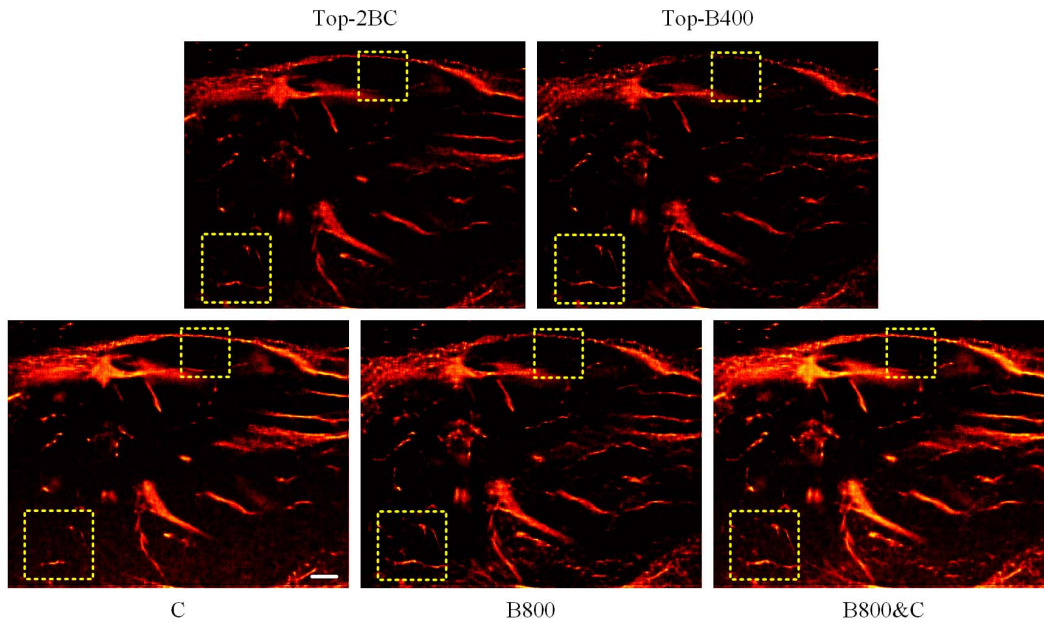


Fig. 14. SR-US images of the *in vivo* rat hindlimb for combined linear and nonlinear US imaging strategies Top-2BC, Top-B400, C, B800, and B800&C utilizing 2400 frames. Insets show magnification of selected shallow and deep microvessels. Shallow vessels and deeper vessels (in boxed areas) are depicted with qualitatively different levels of detail. Scale bar is 2 mm. All figures are at the same scale.

FDA approved US contrast agents are polydisperse, they have different size distributions [45]. More specifically, Definity (Lantheus Medical Imaging) is an MB contrast agent that has a markedly smaller diameter than Optison (GE Healthcare, Inc., Marlborough, MA). Further, the choice of imaging strategy could influence the choice of contrast agent to maximize CTR.

Another limitation of our simulation environment was the use of idealized waveforms for transmit US pulses. Real systems struggle to form ideal waveforms that will cancel well in tissue [47], as seen in Fig. 2. The challenges include making a perfect inversion of the full-amplitude pulse, producing exactly a half-amplitude signal, and accurately controlling the timing between different US pulses. These challenges lead to rise–fall time mismatches, gain error, and phase error. In early investigations of the impact of such errors, simulations show that waveform nonidealities increase the CTR of MBs to nonlinear US pulse sequences. This is understood to be the effect of combining US signals having phase differences, similar to the way the phase of the half-amplitude US pulse response leading the full amplitude response [Fig. 4(c)] enhances nonlinear compositions. The nonidealities of the transmitted US pulse waveforms used in imaging may explain in part why the *in vitro* results exhibited a higher level of MB detection for AMPI-3, while simulations predicted a higher CTR for AM.

The *in vitro* studies suggested that linear US pulsing and different CPS pulsing strategies detected different MB subpopulations. These results from the use of different US pulsing techniques can be compared as they are based on the software combination of the same four basis pulses that were separated by 140 μ s. Thus, for any particular frame in time, the frame for each US imaging strategy shares common information and makes direct comparison valid. Differences in the number of MB detections for a given frame under various strategies may arise as a result from the 140- μ s time

difference between the various basis pulses transmitted for each. During that time interval, MBs may move in the pulsatile flow, or there may be slight MB dissolution, changing the likelihood of detection. The pairing of MBs from the identical half-amplitude pulses separated by $3 \times 140 \mu$ s, revealed a 20% increase the number of distinct MBs that were detected. Thus, 20% is an estimate of the variation in distinct MB detections due to timing variation in this study. While there may be multiple reasons for differences in detection above this 20%, we can infer from the simulation results that levels above this may be due to subpopulations of MBs differing in diameter or position. We plan to study monodisperse MBs moving at constant velocities at various depths in an *in vitro* environment to help validate this hypothesis in future work.

While the use of B-mode US imaging was found to detect a greater number of MBs overall, it appeared to have a higher false-positive rate in areas associated with a strong tissue signal. Conversely, nonlinear US imaging strategies appeared to have a higher false-positive rate arising from noise that is detected as a MB, consistent with previous findings [42]. This is also in agreement with the higher SNR of B-mode US imaging and the lower SNR of nonlinear US imaging, both of which are depth-dependent. The signals for MBs, tissue, and noise, overlap across a range of SVs in the SVD domain. The choice of SV threshold for each US imaging strategy is a compromise between background tissue signal rejection and MB detection [42]. With the greater tissue rejection inherent in nonlinear US imaging, it is less of a problem to reject tissue, although the MB and noise signals appear to overlap more in the decomposition. Additional tissue clutter filtering or noise reduction algorithms could help improve the quality of the final images in each case.

Measured by image intensity saturation from the vessel ROI, the highest *in vitro* MB detection rates for a single imaging

strategy was found with use of B-mode US imaging and for nonlinear US imaging with AMPI-3. Similarly, *in vivo* SR-US images having the qualitatively highest levels of microvascular details were from the AMPI-3-based US images at shallow depths and B-mode US at greater depths. This is understood to come from the higher SNR levels for the CPS compositions at shallower depths due to increasing electronic noise from beamforming and the application of TGC with depth. There seems to be a minimum required SNR level above which the CTR will determine whether a MB is detected. This is understood as SNR influencing whether a MB is distinguished from noise, and CTR influencing whether a MB is distinguished from tissue. For B-mode US imaging, MB detection is limited by tissue suppression, and for nonlinear US imaging strategies, MB detection is limited by the amount of noise present. Further, these observations may favor the detection of certain MB subpopulations based on diameter and other characteristics.

It was shown that the AMPI-3 pulse sequence produced a slightly higher *in vitro* image saturation level and *in vivo* SR-US image quality when compared to the other CPS compositions. This differs from the RPM-kW simulation results, which found slightly higher CTR_R levels using AM pulsing and a theoretically higher MB detection rate. This is consistent with some, but not all *in vitro* studies [27], [34]. For example, it has been shown using Definity MBs that AMPI-3 had a higher CTR than AM or PI [27]. In earlier work, we found the same superiority of the AMPI-3 sequence [35]. If a simple power series relationship is assumed for the MB response to pulsed US, the AMPI-3 sequence retains more second and third-order energy (from terms in the expansion raised to the second and third powers) at the fundamental frequency than either AM or PI [27], and may help interpret this result.

There was an expectation that the RPM model might give a more accurate understanding of the MB acoustic response than the simple power series expansion. This was presumed due to a more complex modeling of second order differential equation of the MB with many parameters characterizing the gas interior, the shell of the MB, and the surrounding environment. While this may be partially true, we found that transmit pulse distortions measured from the actual US system did change the relative performance of the pulsing strategies when modeled as inputs to the RPM simulation. These waveform distortions were attributed to both the US scanner and water tank used for measurement. Due to a higher power law attenuation exponent in water compared to tissue, nonlinear propagation of US has a disparate impact on the frequency content of the waveform [48], and the distortion in water is larger than tissue. As waveform distortions are difficult to model precisely, we suggest the RPM results be viewed as comparative rather than absolute, and the overall trends of the results across MB diameters be considered. The trend that held in simulations with a variety of MB shell parameters and waveform nonidealities was that the CTR for AM, PI, and AMPI, differed from B-mode US having higher CTR at smaller diameters and increasing CTR with MB motion.

Further, this study evaluated PI-2 at the fundamental US frequency rather than using the more common harmonic

imaging method when using a PI sequence. There is more energy in the second harmonic band than at the fundamental frequency for PI. This may partially explain why a lower CTR_R for PI was found in this study.

Multipulse sequences are inherently sensitive to any MB motion. The time between US pulses, here 140 μ s, leads to a phase difference in the backscattered US signals and reduces cancellation. Additionally, three pulse sequence strategies, AMPI-3 and AM-3, may benefit from the time averaging of the two half-amplitude US pulses. This time averaging will tend to reduce noise and raise SNR and benefit three pulse sequences over two US pulse sequences. Three pulses were the maximum considered in order to limit the FR reduction. As the implementation of multiple pulsing in hardware is efficient, an increase in computation time with more US pulse waveforms was less of a concern.

Higher FRs in SR-US supports imaging of faster vascular flows and improves MB tracking under slower flow conditions. However, there is a tradeoff between imaging time and spatial resolution that varies with vessel size [49]. More specifically, it was shown that for slower flow, higher US FRs were not required in order to localize at least one MB per high resolution pixel [29]. For example, with the commonly used 500-Hz FR for SR-US imaging and a depth of 5 cm, flow slower than 40 mm/s would not benefit from a higher FR. Microvascular flow (of 1–10 mm/s) falls within this limit. An FR of 500 Hz allows 30 total pulses per frame (number of angles times number of pulse waveforms). A two or three pulse waveform (e.g., +1.0, -0.5, and -0.5) could be supported with up to ten angles for compounding in this scenario. At 1000 Hz, two or three pulse waveforms could be supported with up to five angles for compounding. In applications with higher flow rates, nonlinear US imaging strategies may be more limited.

While combining linear and nonlinear US pulsing strategies may decrease image acquisition time, a drawback is the increased processing burden. For the fastest processing, a single imaging method would be appropriate and based on the results of this study, B-mode US imaging would lead to the highest number of MBs detected. The results for B800, which is not identical to 800-Hz B-mode US imaging because it uses amplitude and phase modulated pulses, suggests increasing the FR for B-mode US also detects commensurately more MBs. It is unclear how the use of amplitude and phase modulated pulses has confounded this comparison. Of note, we did find some difference in SR-US images formed with the inverted full-amplitude pulse (i.e., B-mode) compared with B-mode US due to a greater number of MB localizations. We suspect that nonlinear aspects of the MB response have in part increased the number of MB detections with B800 as compared to a true B-mode US imaging at 800-Hz strategy.

For reduced US image acquisition time, results presented herein suggests that a good compromise between microvessel visualization and processing rate burden is the Top-2BC strategy, which combines B-mode US and AMPI-3 imaging. The Top-B400 purely linear processing strategy has the same processing rate burden of Top-2BC, but with slightly different qualitative results on *in vivo* US imaging, and also represents a good compromise. If processing time were of primary

importance, a single strategy based on B-mode US imaging at a higher FR of 800 Hz would be predicted to have a similar image acquisition time as the B800 combined strategy. If processing time was less of a concern, the most computationally intensive technique considered here, that of the combined B800&C strategy could accelerate image acquisition time of a strategy based purely on FR (i.e., B800).

Conventional methods for SR-US processing such as described here (see Fig. 3) would execute the processing steps from raw data to SR-US image formation for each US imaging technique and then compound the results. The MB detection and localization steps often require minutes to process an image stack. Novel methods for MB detection and localization utilizing deep learning methods offer promise for achieving *in vivo* SR-US processing times of milliseconds per frame while avoiding the need to acquire an image stack prior to processing to begin [50]–[52]. Such a strategy would provide real-time feedback during the SR-US image formation process akin to conventional image persistence functionality. In the future, the additional computational burden needed for combining a linear and nonlinear US strategy may be quite low with the use of deep learning or other image processing methods [53], [54].

Noteworthy limitations of this study include its focus on a single US platform equipped with a transducer operating at only one center frequency, as well as the study of only one particular MB contrast agent. As the nonlinear MB response decreases with higher frequencies, larger doses of small diameter MBs are used to improve imaging at higher frequencies [47]. Our simulation results suggest a population of smaller MB would have a higher CTR using a nonlinear US imaging strategy and would improve imaging at higher frequencies, an area for future study.

V. CONCLUSION

This study demonstrated that a combination of linear and nonlinear US imaging strategies for SR-US could increase the number of interframe MBs detections and reduce image acquisition time by combining the results from linear B-mode US and nonlinear AMPI-3 imaging. This scheme may be advantageous for use during SR-US imaging of tissue microvascular structures at high FRs including those in the range of 500–1000 Hz. The technique is generalizable to CEUS imaging and may have utility in quantitative techniques that would benefit from maximizing the number of MBs detected per US imaging frame. The simulation results suggest the CTR for MB subpopulations (differing in diameter and motion) was determined by the particular US pulsing technique. This was reflected during *in vitro* comparisons of MB localizations made in overlapping image frames. This insight was made possible by the near simultaneous use and evaluation of a set of basis US pulses that were combined in software. This methodology is broadly applicable to general studies of US pulsing techniques.

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Katherine G. Brown (Student Member, IEEE) received the B.S.E.E. degree from Stanford University, Stanford, CA, USA, in 1983, the M.S.E.E. degree from the University of Washington, Seattle, WA, USA, in 1996, the MBA degree from Southern Methodist University, Dallas, TX, USA, in 2008, and the Ph.D. degree in biomedical engineering from the Dr. Kenneth Hoyt's Ultrasound Imaging and Therapy Laboratory, The University of Texas at Dallas, Richardson, TX, USA, in 2021.

She is currently an Assistant Professor with the Department of Bioengineering, The University of Texas at Dallas. She spent nearly 30 years in industry in engineering and management roles, including Boeing, Seattle, WA, USA, and Siemens Ultrasound, Issaquah, WA, USA, and 19 years at Texas Instruments, Dallas. Her research focuses on super-resolution ultrasound imaging in cancer.



Kenneth Hoyt (Senior Member, IEEE) received the B.S. degree in electrical engineering and the M.S. and Ph.D. degrees in biomedical engineering from Drexel University, Philadelphia, PA, USA, in 2001, 2004, and 2005, respectively, and the M.B.A. degree from the School of Business, Drexel University, Philadelphia, PA, USA, in 2011.

He did a postdoctoral fellowship at the Department of Electrical and Computer Engineering, University of Rochester, Rochester, NY, USA.

He was a Faculty Member with the Department of Radiology, University of Alabama at Birmingham (UAB), Birmingham, AL, USA, from 2008 to 2015. He is currently an Associate Professor with the Department of Bioengineering, The University of Texas at Dallas, Richardson, TX, USA, and the Department of Radiology, University of Texas Southwestern Medical Center, Dallas, TX. His research focuses on the development of novel ultrasound imaging strategies for improved human disease management (e.g., cancer and diabetes).

Dr. Hoyt was elected a fellow of the American Institute of Ultrasound in Medicine (AIUM) in 2014.